

CS310 Natural Language Processing

Assignment 1 (written)

Total points: 10

Consider a softmax layer of output size k ($k > 1$). Within unit i ($i=1, \dots, k$), the linear before activation is z_i , (also called logits), and the output after activation is o_i . We have the following equation for forward propagation:

$$o_i = \frac{\exp(z_i)}{\sum_{j=1}^k \exp(z_j)}$$

The cross-entropy loss between an output prediction $\hat{y} = [o_1, \dots, o_k]$ and the ground truth label $y = [0, \dots, 1, \dots, 0]$ (one-hot encoded) is defined as:

$$L_{CE} = - \sum_{i=1}^k y_i \log(\hat{y}_i) = - \sum_{i=1 \dots k} y_i \log o_i$$

1) Show that when $k=2$, the loss is equivalent to binary cross-entropy (BCE) loss $L_{BCE} = -y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i)$

(4 points, show your math derivation to get the credits)

2) What is the gradient w.r.t the i th unit $\frac{\partial L}{\partial z_i}$?

(6 points, show your math derivation to get the credits)